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ROUND TRIPS AND RESTARTS

- Where $T^m(t)$ and $R^m(t)$ are the total restarts/round trips on machine m by time t

$$T^m(t) = \sum_i 1[T^m_{i,+} \leq t], \qquad R^m(t) = \sum_i 1[T^m_{i,-} \leq t]$$

- Goal is to maximise the number of restarts or equivalently round trips!

$$T(t) = \sum_m T^m(t), \qquad R(t) = \sum_m R^m(t)$$

- Both are equivalent for large t

$$R(t) \leq T(t) \leq R(t) + N$$

- Round-trip and restarts are easy to compute/track from the index process

- Act proxy for the number of unique sources of reference noise in the target chain

- For multi-modal problems, a replica often explores at most one target mode between round-trip

- Due to complex interactions it is hard to analyse in general

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INDEX PROCESS WEAKLY INTERACTIVE